<u>Problems on Gamma and Weibull Distribution</u>

1)

If a company employes n sales persons, its gross sales in thousands of rupees may be regarded as a RV having an Erlang distribution with $\lambda = \frac{1}{2}$ *and* $k = 80\sqrt{n}$*. If the sales cost is Rs 8000 per salesperson, how many salespersons should the company employ to maximise the expected profit?*

2)

Consumer demand for milk in a certain locality, per month, is known to be a general Gamma (Erlang) RV. If the average demand is a litres and the most likely demand is b litres ($b < a$)*, what is the variance of the demand?*

3)

If the service life, in hours, of a semiconductor is a RV having a Weibull distribution with the parameters $\alpha = 0.025$ and $\beta = 0.5$,

(i) how long can such a semiconductor be expected to last? and

(ii) what is the probability that such a semiconductor will still be in operating condition after 4000 h?

4)

Find the probability of failure-free performance of roller-bearings over a period of 10^4 h if the life expectancy of the bearings is defined by Weibull distribution with parameters $\alpha = 10^{-7}$ and $\beta = 1.5$.

[**Hint:** P(failure-free performance over a period t) = P(the component does not fail in $(0, t)$] = $P(T \geq t)$, where T is the life expectancy or time to failure of the component.]